

理论计算机科学导引

Lecture Notes

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Chapter 1

引子.

function
specification
what?

$\Leftrightarrow$

Program
implementation
how

定义1. Binary String.

字符集 $\Sigma = \{0, 1\}$. (可推广)

$$\Sigma^n = \{(a_1, \dots, a_n) \mid a_i \in \Sigma\}.$$

$\downarrow$
Binary String

$\Sigma^0 = \{\epsilon\}$. 空串.

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \dots = \bigcup_{n \geq 0} \Sigma^n$$

x 为 y 前缀: $y = xz$. $\exists z$.

定义2: Encoding.

$E: A \rightarrow \{0, 1\}^*$. 单射.

例2.1: $N \times N \rightarrow \{0, 1\}^*$.

$(a, b) \rightarrow E(a)E(b) \quad \times \quad (E(1,6) = E(3,2)).$
prefix 冲突.

$\therefore \exists \lambda$ prefix-free.

定义3. prefix-free Encoding

$E(x)$ 不为 $E(x')$ 前缀, $\forall x \neq x'$

有: $E: A \rightarrow \{0,1\}^*$, prefix-free

欲得: $\bar{E}: A^* \rightarrow \{0,1\}^*$, one-to-one.

思考: $(a_1, \dots, a_n) \in A^*$

令: $\bar{E}(a_1, \dots, a_n) = \begin{cases} E(a_1) \dots E(a_n), & n \neq 0 \\ \epsilon, & n = 0 \end{cases}$
(自然定义).

为什么对?

引理1: $\bar{E}$ 为单射.

证: 若 $\exists (a_1, \dots, a_p) \neq (b_1, \dots, b_q).$

但 $E(a \dots) = E(b \dots).$

find first $i : a_i \neq b_i$

则 $E(a_1) \dots E(a_p) = E(b_1) \dots E(b_q)$ ($a_1 \sim a_{i-1}$ 已匹配).

$\therefore E(a_i)$ 与 $E(b_i)$ 有前缀关系.

矛盾!

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引理 2.

若 $\exists$ encoding $E: A \rightarrow \{0,1\}^*$

则 $\exists$ prefix-free $E': A \rightarrow \{0,1\}^*$

s.t. $|E'(a)| \leq 2|E(a)| + 2, \forall a.$

证:

$E(a) = 010, E'(a) = 001100 + 01$ (显然 one-to-one)

此时: 若 $E(x)$ 为 $E(y)$ 前缀, 则 $\text{len}(E(x)) \leq \text{len}(E(y)).$

若 $\text{len}(E(x)) < \text{len}(E(y))$, 必有 "01" 匹配 "00"/"11". 矛盾.

$\therefore \text{len}(E(x)) = \text{len}(E(y))$

$\therefore E(x) = E(y). \quad x = y.$

#.

思考: $|E'(a)| \leq |E(a)| + O(\log |E(a)|).$?

定理1:

若 $\exists$ one-to-one $E: A \rightarrow \{0,1\}^*$.

则 $\exists$ one-to-one $E': A^* \rightarrow \{0,1\}^*$.

证: 由引理1.2. 立得.

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定义4. 可数. (至多可数).

(1). 有限

(2). A 与 N 双射 $\Downarrow$.

(3). $\exists$ one-to-one $f: A \rightarrow N$.

(4). $\exists$ onto func. $g: N \rightarrow A$.

引理3: $\{0,1\}^*$ 可数.

(有限集的可列并可数).

定理2: Countable $\Leftrightarrow$ 可 Encoding.

(比较 trivial 略了...)

定义5. Problem

$$f: \{0,1\}^* \rightarrow \{0,1\}^*$$

定理3: Problem is Uncountable.

Cantor 对角化.
